

Associative Memory in Fast-Weight Architectures: The Geometry of Recurrent Interference

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Target Reader: Machine learning practitioners studying linear recurrent memory limits.

The Bottleneck of Linear Recurrence

Autoregressive Transformers rely on Softmax Attention, which requires storing all past Key-Value (KV) vectors ($\mathcal{O}(N)$ memory per step). Over long sequences, this cache strains hardware memory bandwidth.

Linear recurrent models and fast-weight architectures (Schlag et al., 2021; Gu & Dao, 2023; Beck et al., 2024) compress context into a fixed-size state matrix $S_t \in \mathbb{R}^{d \times d}$ updated via outer products:

$$S_t = S_{t-1} + v_t k_t^T, \quad S_0 = \mathbf{0}$$

Reading from memory uses matrix-vector multiplication with query q :

$$y = Sq = \sum_{i=1}^N v_i (k_i^T q)$$

When probing for stored key k_j ($q = k_j$), retrieval decomposes into target signal and cross-talk interference:

$$y_j = \underbrace{v_j (k_j^T k_j)}_{\text{Target Signal}} + \underbrace{\sum_{i \neq j} v_i (k_i^T k_j)}_{\text{Cross-Talk Interference}}$$

The Mechanism of Interference

Our central claim is: **When stored keys are not orthogonal, a linear fast-weight state adds non-target contributions to retrieval; in our synthetic setup, increasing controlled key overlap increases measured cross-talk.**

Under orthonormal keys ($k_i^T k_j = 0$ for $i \neq j$), cross-talk vanishes, yielding exact reconstruction $y_j = v_j$. However, when unit keys in $\mathbb{R}^d$ are not orthogonal, off-diagonal inner products inject non-target values additively into the readout. Because the recurrent state is an un-gated outer-product sum, query evaluation distributes linearly across all stored items, scaling each non-target value by its key projection.

Our benchmark sweeps (50 Monte Carlo trials per configuration at $d = 8$) measured retrieval degradation across controlled key correlation ρ :

- Orthogonal Baseline** ($\rho = 0.0, N = 4$): Mean Cosine Similarity = 1.0000, Mean L2 Error = 0.0000.
- Moderate Overlap** ($\rho = 0.45, N = 4$): Mean Cosine Similarity drops to 0.7647, Mean L2 Error rises to 0.8475.
- High Load** ($N = 12 > d = 8, \rho = 0.35$): Mean Cosine Similarity drops to 0.5764, Mean L2 Error rises to 1.3590.

Across the sweep, higher controlled key overlap is associated with higher measured cross-talk, with small stochastic variation between configurations. The linear state has rank bounded by d ($\text{rank}(S) \leq \min(d, N)$). When $N > d$, nonzero keys in $\mathbb{R}^d$ cannot all be mutually orthogonal, producing non-zero projections that increase measured error.

Stress Testing: High-Load Regime

In an adversarial test ($d = 6, N = 10, \rho = 0.75$), cumulative interference terms ($\|\sum_{i \neq j} v_i (k_i^T k_j)\|$) exceed the target signal ($\|v_j\|$). The retrieved vector aligns with the dominant cluster of interfering keys rather than the target value, severely distorting recall.

Connection to Dragon Hatchling (BDH)

Dragon Hatchling (**BDH**; Kosowski et al., 2025) explores a brain-inspired alternative using sparse positive activations (ReLU/Top-K) and local synaptic plasticity. Under sparse support, restricting active connections alters feature

interactions compared to dense linear superposition. **BDH-CQ** (Engdahl et al., 2026) extends this to in-context demonstration learning ($X \rightarrow Y$) via recurrent latent computation.

Our non-negative sparse teaching abstraction changes which interactions contribute to the recurrent state, but it should not be interpreted as evidence that the full BDH architecture eliminates cross-talk. Under finite dimension d , memory representation remains fundamentally constrained by finite-dimensional geometric bounds. Understanding the precise trade-off between synaptic sparsity and representational capacity across non-uniform natural language token distributions remains an open research frontier.

Primary References

1. **Schlag, I., Irie, K., & Schmidhuber, J. (2021).** *Linear Transformers Are Secretly Fast Weight Programmers*. ICML. [arXiv:2102.11174](https://arxiv.org/abs/2102.11174).
2. **Beck, M., et al. (2024).** *xLSTM: Extended Long Short-Term Memory*. NeurIPS. [arXiv:2405.04517](https://arxiv.org/abs/2405.04517).
3. **Gu, A., & Dao, T. (2023).** *Mamba: Linear-Time Sequence Modeling*. [arXiv:2312.00752](https://arxiv.org/abs/2312.00752).
4. **Sun, Y., et al. (2024).** *Learning to (Learn at Test Time): RNNs with Expressive Hidden States*. [arXiv:2407.04620](https://arxiv.org/abs/2407.04620).
5. **Kosowski, A., et al. (2025).** *The Dragon Hatchling: Missing Link between Transformer and Brain*. [arXiv:2509.26507](https://arxiv.org/abs/2509.26507).
6. **Engdahl, B., et al. (2026).** *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning*. [arXiv:2608.09888](https://arxiv.org/abs/2608.09888).